

| Tool Version : Vivado v.2025.2 (win64) Build 6299465 Fri Nov 14
19:35:11 GMT 2025
| Date : Sat Jul 25 12:16:15 2026
| Host : AK_LAPTOP running 64-bit major release (build
9200)
| Command : report_utilization -file
cordic_top_utilization_synth.rpt -pb
cordic_top_utilization_synth.pb
| Design : cordic_top
| Device : xc7a15tcp236-2
| Speed File : -2
Design State : Synthesized

Utilization Design Information

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1. Slice Logic

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Site Type	Used	Fixed	Prohibited	Available
Util%				
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Slice LUTs*	127	0	0	10400
1.22				
LUT as Logic	127	0	0	10400
1.22				
LUT as Memory	0	0	0	9600
0.00				
Slice Registers	57	0	0	20800
0.27				
Register as Flip Flop	57	0	0	20800
0.27				
Register as Latch	0	0	0	20800
0.00				
F7 Muxes	0	0	0	16300
0.00				
F8 Muxes	0	0	0	8150
0.00				
Unique Control Sets	3		0	8150
0.04				
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* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

Warning! LUT value is adjusted to account for LUT combining.

Warning! For any ECO changes, please run place_design if there are unplaced instances

** Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more information regarding control sets.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	—	—	—
0	—	—	Set
0	—	—	Reset
0	—	Set	—
0	—	Reset	—
0	Yes	—	—
0	Yes	—	Set
0	Yes	—	Reset
1	Yes	Set	—
56	Yes	Reset	—

2. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	25	0.00
RAMB36/FIFO*	0	0	0	25	0.00
RAMB18	0	0	0	50	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	2	0	0	45	4.44
DSP48E1 only	2				

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	101	0	0		
Bonded IPADs	0	0	0		
Bonded OPADs	0	0	0		
PHY_CONTROL	0	0	0		
PHASER_REF	0	0	0		
OUT_FIFO	0	0	0		
IN_FIFO	0	0	0		
IDELAYCTRL	0	0	0		
IBUFDS	0	0	0		

1	GTPE2_CHANNEL	0	0	0
2	0.00			
	PHASER_OUT/PHASER_OUT_PHY	0	0	0
20	0.00			
	PHASER_IN/PHASER_IN_PHY	0	0	0
20	0.00			
	IDELAYE2/IDELAYE2_FINEDELAY	0	0	0
250	0.00			
	IBUFDS_GTE2	0	0	0
2	0.00			
	ILOGIC	0	0	0
106	0.00			
	OLOGIC	0	0	0
106	0.00			
+-----+-----+-----+-----+				
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5. Clocking

Site	Type	Used	Fixed	Prohibited	Available	Util%
BUFGCTRL		1	0	0	32	3.13
BUFIO		0	0	0	20	0.00
MMCME2_ADV		0	0	0	5	0.00
PLLE2_ADV		0	0	0	5	0.00
BUFMRCE		0	0	0	10	0.00
BUFHCE		0	0	0	72	0.00
BUFR		0	0	0	20	0.00
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6. Specific Feature

Site Type	Used	Fixed	Prohibited	Available	Util%
BSCANE2	0	0	0	4	0.00
CAPTUREE2	0	0	0	1	0.00
DNA_PORT	0	0	0	1	0.00
EFUSE_USR	0	0	0	1	0.00
FRAME_ECCE2	0	0	0	1	0.00
ICAPE2	0	0	0	2	0.00
PCIE_2_1	0	0	0	1	0.00
STARTUPE2	0	0	0	1	0.00
XADC	0	0	0	1	0.00

7. Primitives

Ref Name	Used	Functional Category
LUT5	66	LUT
FDRE	56	Flop & Latch
IBUF	52	IO
OBUF	49	IO
LUT6	40	LUT
LUT3	23	LUT
CARRY4	12	CarryLogic
LUT4	9	LUT
LUT2	4	LUT
DSP48E1	2	Block Arithmetic
LUT1	1	LUT
FDSE	1	Flop & Latch
BUFG	1	Clock

8. Black Boxes

+-----+-----+	
Ref Name	Used
+-----+-----+	

9. Instantiated Netlists

+-----+-----+	
Ref Name	Used
+-----+-----+	